Results for algebra2005

Current results grouped by recommendation-graph parameters | Snapshot: 2026-10-06 08:40 CST

Each group uses the same six model variants, N = 5, 10, ..., 50, and four Ep response thresholds.

Bold = best; underline = second best by unrounded value. Rankings require all six models.

Parameter group	Completed models	Run status
1. theta_m=0.8, theta_f=0.8, rec=10	**6/6**	completed
2. theta_m=0.5, theta_f=0.2, rec=10	**6/6**	completed
3. theta_m=0.7, theta_f=0.8, rec=10	**6/6**	completed
4. theta_m=0.5, theta_f=0.8, rec=10	**6/6**	completed
5. theta_m=0.8, theta_f=0.7, rec=10	**6/6**	completed
6. theta_m=0.8, theta_f=0.5, rec=10	**6/6**	completed
7. theta_m=0.8, theta_f=0.8, rec=30	**6/6**	completed
8. theta_m=0.8, theta_f=0.8, rec=50	**6/6**	completed

Metric source

Data_Fin_New/_sensitivity_runs/<variant>/<dataset>/sensitivity_seed2024/
SemanticConvE_<model>/seed2024/eval/metrics.json

Group 1: theta_m=0.8, theta_f=0.8, rec=10

6/6 models completed | completed | ranked across six models

(a) Adaptability (Ada)

N	2CKG4ER	No-mastery	No-forgetting	No-student-feature	No-exercise-feature	ID-based only
5	0.748	0.682	**0.788**	0.693	0.748	0.744
10	0.735	0.650	**0.766**	0.668	0.728	0.729
15	0.734	0.621	**0.744**	0.644	0.729	0.714
20	**0.726**	0.588	0.718	0.626	**0.726**	0.693
25	0.697	0.544	**0.703**	0.608	0.701	0.668
30	0.683	0.521	**0.701**	0.588	0.672	0.649
35	0.652	0.504	**0.673**	0.571	0.652	0.625
40	**0.645**	0.488	0.644	0.557	0.629	0.603
45	**0.622**	0.464	0.616	0.541	0.621	0.592
50	0.600	0.444	**0.601**	0.526	0.597	0.578

(b) Novelty (NOV)

N	2CKG4ER	No-mastery	No-forgetting	No-student-feature	No-exercise-feature	ID-based only
5	0.989	0.982	**0.994**	0.990	0.989	0.986
10	0.988	0.983	**0.992**	0.988	0.987	0.988
15	0.988	0.982	**0.990**	0.989	0.987	0.986
20	0.986	0.981	0.987	**0.988**	0.986	0.985
25	0.986	0.980	**0.988**	0.987	0.985	0.985
30	0.986	0.980	**0.986**	0.986	0.986	0.986
35	0.986	0.980	**0.988**	0.985	0.986	0.986
40	0.986	0.979	**0.988**	0.985	0.987	0.986
45	0.987	0.978	**0.987**	0.985	0.987	0.986
50	0.986	0.977	**0.988**	0.984	0.987	0.986

Seed 2024; single-run values. No across-seed standard deviation is available.

Group 1: theta_m=0.8, theta_f=0.8, rec=10

6/6 models completed | completed | ranked across six models

(c) Ep, response threshold = 0.4

N	2CKG4ER	No-mastery	No-forgetting	No-student-feature	No-exercise-feature	ID-based only
5	-0.152	**0.040**	-0.104	-0.009	-0.026	-0.018
10	**0.145**	0.098	-0.057	0.060	0.053	0.021
15	0.073	0.111	**0.192**	0.072	-0.009	0.072
20	0.098	0.166	**0.307**	0.101	0.004	0.128
25	0.269	0.181	0.299	0.141	**0.304**	0.185
30	**0.338**	0.163	0.225	0.131	0.283	0.250
35	0.316	0.086	**0.399**	0.144	0.306	0.280
40	0.241	0.068	**0.330**	0.150	0.151	0.208
45	0.120	0.044	0.089	0.139	**0.198**	0.139
50	0.293	0.072	-0.007	0.128	**0.399**	0.178

(d) Ep, response threshold = 0.5

N	2CKG4ER	No-mastery	No-forgetting	No-student-feature	No-exercise-feature	ID-based only
5	-0.235	**0.010**	-0.110	-0.034	-0.033	-0.029
10	**0.140**	0.079	-0.153	0.028	0.052	0.000
15	0.073	0.106	**0.174**	0.052	-0.009	0.052
20	0.098	0.165	**0.306**	0.084	-0.125	0.111
25	0.185	0.179	**0.299**	0.127	0.247	0.173
30	**0.318**	0.153	0.225	0.120	0.263	0.239
35	0.310	0.058	**0.399**	0.133	0.301	0.275
40	0.237	0.044	**0.330**	0.138	-0.004	0.202
45	0.039	0.024	0.089	0.121	**0.150**	0.121
50	0.277	0.057	-0.007	0.108	**0.389**	0.161

Seed 2024; single-run values. No across-seed standard deviation is available.

Group 1: theta_m=0.8, theta_f=0.8, rec=10

6/6 models completed | completed | ranked across six models

(e) Ep, response threshold = 0.6

N	2CKG4ER	No-mastery	No-forgetting	No-student-feature	No-exercise-feature	ID-based only
5	-0.347	-0.104	-0.313	-0.138	-0.104	**-0.093**
10	**0.020**	-0.053	-0.410	-0.110	-0.036	-0.090
15	-0.042	**0.003**	-0.206	-0.084	-0.036	-0.048
20	-0.013	0.144	**0.268**	-0.032	-0.311	-0.038
25	0.062	0.170	**0.221**	0.021	0.038	0.030
30	**0.283**	0.141	0.182	0.016	0.122	0.170
35	**0.296**	0.041	0.214	0.044	0.274	0.234
40	0.132	0.009	**0.299**	0.053	-0.021	0.165
45	0.020	-0.003	-0.016	0.034	**0.139**	0.078
50	0.271	0.025	-0.040	0.021	**0.384**	0.100

(f) Ep, response threshold = 0.7

N	2CKG4ER	No-mastery	No-forgetting	No-student-feature	No-exercise-feature	ID-based only
5	-0.428	-0.202	-0.409	-0.246	-0.286	**-0.202**
10	-0.206	**-0.182**	-0.436	-0.251	-0.300	-0.282
15	-0.317	**-0.190**	-0.231	-0.254	-0.368	-0.265
20	-0.307	-0.022	**0.057**	-0.229	-0.407	-0.233
25	-0.164	**0.043**	-0.060	-0.194	-0.004	-0.200
30	**0.211**	0.079	-0.006	-0.208	-0.084	-0.091
35	0.055	-0.020	**0.178**	-0.191	-0.008	0.020
40	-0.131	-0.084	**0.130**	-0.170	-0.239	-0.049
45	-0.302	**-0.103**	-0.344	-0.184	-0.257	-0.203
50	0.137	-0.083	-0.432	-0.197	**0.279**	-0.195

Group 2: theta_m=0.5, theta_f=0.2, rec=10

6/6 models completed | completed | ranked across six models

(a) Adaptability (Ada)

N	2CKG4ER	No-mastery	No-forgetting	No-student-feature	No-exercise-feature	ID-based only
5	0.633	0.201	**0.708**	0.574	0.672	0.671
10	0.667	0.315	**0.705**	0.550	0.645	0.660
15	0.652	0.283	**0.709**	0.541	0.638	0.646
20	0.654	0.297	**0.707**	0.529	0.632	0.637
25	0.647	0.286	**0.711**	0.521	0.640	0.618
30	0.623	0.281	**0.678**	0.514	0.632	0.599
35	0.631	0.281	**0.657**	0.509	0.618	0.587
40	0.627	0.274	**0.648**	0.502	0.605	0.582
45	0.609	0.268	**0.630**	0.499	0.598	0.574
50	0.597	0.262	**0.610**	0.494	0.594	0.565

(b) Novelty (NOV)

N	2CKG4ER	No-mastery	No-forgetting	No-student-feature	No-exercise-feature	ID-based only
5	0.983	0.970	**0.989**	0.979	0.981	0.988
10	0.982	0.974	**0.987**	0.982	0.979	0.987
15	0.985	0.974	**0.987**	0.982	0.983	0.984
20	0.985	0.976	0.986	0.984	0.985	**0.986**
25	0.986	0.978	**0.986**	0.985	0.986	0.986
30	0.986	0.977	0.986	0.985	0.986	**0.987**
35	0.987	0.977	0.987	0.985	0.987	**0.988**
40	0.987	0.978	0.987	0.986	0.987	**0.988**
45	0.988	0.978	0.988	0.986	0.987	**0.989**
50	0.988	0.977	0.987	0.986	0.988	**0.989**

Seed 2024; single-run values. No across-seed standard deviation is available.

Group 2: theta_m=0.5, theta_f=0.2, rec=10

6/6 models completed | completed | ranked across six models

(c) Ep, response threshold = 0.4

N	2CKG4ER	No-mastery	No-forgetting	No-student-feature	No-exercise-feature	ID-based only
5	0.115	-0.034	**0.152**	0.054	0.144	0.138
10	0.039	-0.052	0.040	0.080	**0.132**	0.115
15	-0.081	0.135	-0.035	0.083	**0.153**	0.088
20	-0.072	0.021	**0.123**	0.098	-0.026	0.043
25	-0.187	**0.244**	0.146	0.098	-0.108	0.037
30	-0.001	-0.015	**0.148**	0.100	-0.075	0.000
35	0.059	-0.005	**0.254**	0.088	0.070	0.097
40	0.248	**0.259**	0.010	0.099	0.184	0.118
45	0.120	0.197	-0.041	0.126	**0.248**	0.147
50	0.182	0.143	0.194	0.142	**0.284**	0.146

(d) Ep, response threshold = 0.5

N	2CKG4ER	No-mastery	No-forgetting	No-student-feature	No-exercise-feature	ID-based only
5	0.111	-0.034	0.138	0.012	**0.140**	0.132
10	0.036	-0.052	0.038	0.021	**0.120**	0.109
15	-0.082	0.135	-0.036	0.015	**0.148**	0.060
20	-0.072	0.021	**0.122**	0.010	-0.076	-0.033
25	-0.205	**0.244**	0.146	0.029	-0.130	-0.026
30	-0.149	-0.015	0.015	**0.028**	-0.165	-0.098
35	0.015	-0.086	**0.234**	0.016	-0.122	0.029
40	0.216	**0.252**	0.003	0.023	0.084	0.070
45	0.111	0.170	-0.119	0.069	**0.219**	0.122
50	0.179	0.136	0.182	0.096	**0.259**	0.064

Seed 2024; single-run values. No across-seed standard deviation is available.

Group 2: theta_m=0.5, theta_f=0.2, rec=10

6/6 models completed | completed | ranked across six models

(e) Ep, response threshold = 0.6

N	2CKG4ER	No-mastery	No-forgetting	No-student-feature	No-exercise-feature	ID-based only
5	0.101	-0.036	0.014	-0.055	**0.128**	0.106
10	-0.144	-0.054	-0.019	-0.085	-0.018	**0.095**
15	-0.159	**0.113**	-0.048	-0.111	0.104	-0.043
20	-0.259	**0.019**	0.006	-0.118	-0.172	-0.195
25	-0.485	**0.244**	0.126	-0.109	-0.332	-0.220
30	-0.345	-0.016	**0.010**	-0.107	-0.443	-0.327
35	-0.215	-0.118	**0.193**	-0.112	-0.350	-0.249
40	0.016	**0.216**	-0.011	-0.112	-0.223	-0.204
45	-0.185	**0.142**	-0.209	-0.061	-0.045	-0.124
50	0.110	0.127	**0.130**	-0.026	0.113	-0.178

(f) Ep, response threshold = 0.7

N	2CKG4ER	No-mastery	No-forgetting	No-student-feature	No-exercise-feature	ID-based only
5	0.039	-0.114	-0.112	-0.115	**0.087**	-0.019
10	-0.254	-0.263	-0.283	-0.192	-0.132	**-0.012**
15	-0.441	**-0.101**	-0.344	-0.234	-0.175	-0.201
20	-0.458	**-0.159**	-0.253	-0.249	-0.409	-0.337
25	-0.508	**0.212**	-0.282	-0.252	-0.420	-0.347
30	-0.355	**-0.172**	-0.313	-0.246	-0.458	-0.402
35	-0.393	-0.270	**-0.207**	-0.256	-0.357	-0.343
40	-0.229	**0.124**	-0.406	-0.242	-0.262	-0.335
45	-0.322	**0.099**	-0.441	-0.210	-0.153	-0.291
50	-0.213	**0.102**	-0.155	-0.193	-0.092	-0.310

Seed 2024; single-run values. No across-seed standard deviation is available.

Group 3: theta_m=0.7, theta_f=0.8, rec=10

6/6 models completed | completed | ranked across six models

(a) Adaptability (Ada)

N	2CKG4ER	No-mastery	No-forgetting	No-student-feature	No-exercise-feature	ID-based only
5	0.714	0.613	**0.770**	0.686	0.736	0.748
10	0.724	0.609	**0.750**	0.656	0.729	0.735
15	0.721	0.595	**0.742**	0.637	0.722	0.734
20	0.704	0.570	**0.739**	0.618	0.703	0.716
25	0.677	0.547	**0.723**	0.603	0.683	0.693
30	0.647	0.532	**0.695**	0.592	0.651	0.648
35	0.633	0.508	**0.682**	0.580	0.628	0.630
40	0.607	0.482	**0.647**	0.568	0.610	0.602
45	0.595	0.466	**0.617**	0.557	0.588	0.587
50	0.579	0.445	**0.595**	0.544	0.576	0.579

(b) Novelty (NOV)

N	2CKG4ER	No-mastery	No-forgetting	No-student-feature	No-exercise-feature	ID-based only
5	0.986	0.978	**0.994**	0.986	0.988	0.989
10	0.985	0.979	**0.991**	0.985	0.985	0.988
15	0.984	0.979	**0.991**	0.987	0.984	0.988
20	0.984	0.979	**0.990**	0.987	0.984	0.985
25	0.985	0.980	**0.989**	0.986	0.986	0.983
30	0.985	0.979	**0.989**	0.986	0.986	0.984
35	0.986	0.979	**0.987**	0.986	0.986	0.985
40	0.985	0.977	**0.988**	0.986	0.986	0.986
45	0.985	0.977	**0.988**	0.985	0.986	0.986
50	0.985	0.976	**0.989**	0.985	0.985	0.985

Seed 2024; single-run values. No across-seed standard deviation is available.

Group 3: theta_m=0.7, theta_f=0.8, rec=10

6/6 models completed | completed | ranked across six models

(c) Ep, response threshold = 0.4

N	2CKG4ER	No-mastery	No-forgetting	No-student-feature	No-exercise-feature	ID-based only
5	0.028	**0.091**	-0.165	-0.041	-0.046	-0.111
10	0.023	**0.135**	-0.010	0.027	0.000	0.131
15	0.005	**0.181**	0.091	0.081	0.036	0.012
20	0.014	**0.223**	0.159	0.110	0.033	0.022
25	0.130	0.160	0.263	0.130	0.212	**0.315**
30	0.112	0.053	0.238	0.161	0.027	**0.295**
35	0.061	0.051	**0.266**	0.196	0.152	0.173
40	0.063	-0.036	0.138	0.199	0.137	**0.292**
45	0.191	0.033	0.235	0.208	0.257	**0.324**
50	0.208	0.062	**0.238**	0.205	0.180	0.165

(d) Ep, response threshold = 0.5

N	2CKG4ER	No-mastery	No-forgetting	No-student-feature	No-exercise-feature	ID-based only
5	0.008	**0.074**	-0.180	-0.069	-0.063	-0.130
10	0.019	**0.130**	-0.095	-0.031	-0.003	0.130
15	-0.006	**0.180**	0.063	0.033	0.031	0.012
20	0.012	**0.223**	0.149	0.086	0.017	0.015
25	0.125	0.159	0.261	0.113	0.171	**0.313**
30	0.106	0.005	0.238	0.148	0.017	**0.295**
35	0.055	0.027	**0.266**	0.184	0.146	0.173
40	0.060	-0.052	0.138	0.185	0.120	**0.250**
45	0.189	0.016	0.235	0.196	0.247	**0.315**
50	0.200	0.027	**0.238**	0.190	0.176	0.160

Seed 2024; single-run values. No across-seed standard deviation is available.

Group 3: theta_m=0.7, theta_f=0.8, rec=10

6/6 models completed | completed | ranked across six models

(e) Ep, response threshold = 0.6

N	2CKG4ER	No-mastery	No-forgetting	No-student-feature	No-exercise-feature	ID-based only
5	-0.051	**-0.002**	-0.364	-0.170	-0.133	-0.238
10	-0.020	**0.073**	-0.383	-0.155	-0.076	0.050
15	-0.122	**0.166**	-0.342	-0.102	-0.097	-0.031
20	-0.170	**0.220**	-0.298	-0.039	-0.135	-0.259
25	0.036	0.138	-0.073	-0.012	0.082	**0.192**
30	0.042	-0.023	0.141	0.033	-0.011	**0.247**
35	-0.038	-0.009	**0.163**	0.086	0.124	0.152
40	-0.029	-0.095	0.052	0.089	-0.009	**0.110**
45	0.130	-0.040	-0.015	0.112	0.161	**0.281**
50	**0.166**	-0.024	0.121	0.110	0.123	0.143

(f) Ep, response threshold = 0.7

N	2CKG4ER	No-mastery	No-forgetting	No-student-feature	No-exercise-feature	ID-based only
5	-0.160	**-0.114**	-0.413	-0.254	-0.286	-0.370
10	-0.242	**-0.111**	-0.413	-0.274	-0.289	-0.221
15	-0.302	**0.042**	-0.367	-0.257	-0.348	-0.384
20	-0.350	**0.127**	-0.321	-0.233	-0.339	-0.355
25	-0.302	**0.063**	-0.187	-0.214	-0.266	-0.026
30	-0.262	-0.096	-0.161	-0.203	-0.322	**0.031**
35	-0.340	-0.140	**0.023**	-0.153	-0.218	-0.115
40	-0.359	-0.202	-0.087	-0.141	-0.269	**-0.058**
45	-0.205	-0.188	-0.136	-0.134	-0.085	**0.049**
50	-0.091	-0.112	**-0.032**	-0.135	-0.063	-0.100

Seed 2024; single-run values. No across-seed standard deviation is available.

Group 4: theta_m=0.5, theta_f=0.8, rec=10

6/6 models completed | completed | ranked across six models

(a) Adaptability (Ada)

N	2CKG4ER	No-mastery	No-forgetting	No-student-feature	No-exercise-feature	ID-based only
5	0.649	0.642	**0.694**	0.627	0.688	0.666
10	0.662	0.625	**0.705**	0.603	0.654	0.680
15	0.670	0.609	0.703	0.582	0.656	**0.703**
20	0.660	0.574	0.707	0.565	0.656	**0.711**
25	0.639	0.554	**0.693**	0.552	0.637	0.683
30	0.614	0.542	**0.689**	0.539	0.616	0.647
35	0.587	0.511	**0.660**	0.524	0.592	0.622
40	0.575	0.484	**0.646**	0.512	0.577	0.595
45	0.565	0.459	**0.630**	0.501	0.565	0.583
50	0.555	0.442	**0.611**	0.490	0.554	0.553

(b) Novelty (NOV)

N	2CKG4ER	No-mastery	No-forgetting	No-student-feature	No-exercise-feature	ID-based only
5	0.975	0.979	**0.987**	0.984	0.980	0.975
10	0.977	0.982	**0.987**	0.984	0.980	0.980
15	0.981	0.982	**0.987**	0.985	0.981	0.983
20	0.982	0.980	**0.986**	0.985	0.982	0.984
25	0.982	0.981	0.985	**0.986**	0.983	0.985
30	0.983	0.980	0.986	**0.986**	0.983	0.982
35	0.982	0.979	**0.987**	0.986	0.983	0.982
40	0.983	0.977	**0.987**	0.985	0.983	0.984
45	0.983	0.977	**0.988**	0.985	0.983	0.984
50	0.984	0.976	**0.987**	0.984	0.983	0.983

Seed 2024; single-run values. No across-seed standard deviation is available.

Group 4: theta_m=0.5, theta_f=0.8, rec=10

6/6 models completed | completed | ranked across six models

(c) Ep, response threshold = 0.4

N	2CKG4ER	No-mastery	No-forgetting	No-student-feature	No-exercise-feature	ID-based only
5	0.039	0.090	**0.104**	0.023	0.008	-0.007
10	0.001	0.157	**0.170**	0.043	0.008	-0.141
15	0.020	**0.196**	0.149	0.069	0.042	-0.020
20	0.080	**0.232**	0.030	0.079	0.071	0.050
25	**0.209**	0.181	0.027	0.119	0.095	0.018
30	0.060	0.049	0.159	0.126	0.104	**0.256**
35	0.072	0.055	0.113	0.133	0.033	**0.185**
40	0.119	0.033	-0.082	0.166	0.034	**0.259**
45	**0.152**	0.032	-0.057	0.138	0.118	0.135
50	0.078	0.083	0.025	0.142	**0.188**	0.024

(d) Ep, response threshold = 0.5

N	2CKG4ER	No-mastery	No-forgetting	No-student-feature	No-exercise-feature	ID-based only
5	0.006	0.068	**0.091**	-0.004	-0.018	-0.013
10	-0.027	0.149	**0.166**	0.006	-0.014	-0.142
15	0.014	**0.194**	0.149	0.037	0.028	-0.024
20	0.078	**0.230**	0.024	0.044	0.064	0.050
25	**0.209**	0.178	0.026	0.079	0.078	0.017
30	0.060	0.011	0.159	0.090	0.098	**0.256**
35	0.065	0.040	0.080	0.097	0.031	**0.185**
40	0.116	0.022	-0.123	0.123	0.032	**0.259**
45	**0.137**	-0.002	-0.116	0.102	0.117	0.135
50	0.069	0.059	0.011	0.111	**0.168**	0.024

Seed 2024; single-run values. No across-seed standard deviation is available.

Group 4: theta_m=0.5, theta_f=0.8, rec=10

6/6 models completed | completed | ranked across six models

(e) Ep, response threshold = 0.6

N	2CKG4ER	No-mastery	No-forgetting	No-student-feature	No-exercise-feature	ID-based only
5	-0.085	-0.024	**-0.020**	-0.075	-0.111	-0.088
10	-0.135	**0.091**	0.059	-0.082	-0.148	-0.219
15	-0.130	**0.181**	0.137	-0.069	-0.088	-0.303
20	-0.047	**0.226**	-0.001	-0.080	-0.038	-0.332
25	0.138	**0.158**	0.007	-0.055	0.004	-0.057
30	-0.026	-0.019	**0.155**	-0.044	0.012	-0.040
35	-0.063	0.010	0.079	-0.037	-0.086	**0.127**
40	-0.057	-0.013	-0.326	0.003	-0.139	**0.235**
45	**0.065**	-0.075	-0.352	-0.018	-0.015	0.007
50	-0.055	0.008	-0.073	0.006	**0.119**	-0.052

(f) Ep, response threshold = 0.7

N	2CKG4ER	No-mastery	No-forgetting	No-student-feature	No-exercise-feature	ID-based only
5	-0.158	-0.140	-0.155	-0.149	-0.166	**-0.113**
10	-0.268	**-0.095**	-0.170	-0.202	-0.286	-0.383
15	-0.300	**0.051**	-0.077	-0.219	-0.331	-0.422
20	-0.283	**0.126**	-0.328	-0.241	-0.322	-0.386
25	-0.119	**0.099**	-0.370	-0.219	-0.251	-0.386
30	-0.295	**-0.079**	-0.289	-0.214	-0.240	-0.150
35	-0.298	**-0.097**	-0.344	-0.201	-0.296	-0.188
40	-0.241	-0.141	-0.463	-0.180	-0.303	**0.068**
45	-0.206	-0.200	-0.388	**-0.199**	-0.224	-0.232
50	-0.276	**-0.078**	-0.251	-0.189	-0.127	-0.158

Seed 2024; single-run values. No across-seed standard deviation is available.

Group 5: theta_m=0.8, theta_f=0.7, rec=10

6/6 models completed | completed | ranked across six models

(a) Adaptability (Ada)

N	2CKG4ER	No-mastery	No-forgetting	No-student-feature	No-exercise-feature	ID-based only
5	0.748	0.656	**0.790**	0.710	0.745	0.762
10	0.735	0.676	**0.758**	0.705	0.726	0.741
15	0.724	0.679	**0.738**	0.700	0.696	0.730
20	0.709	0.616	0.714	0.684	0.698	**0.726**
25	0.701	0.607	**0.703**	0.666	0.690	0.701
30	0.662	0.576	**0.702**	0.649	0.662	0.671
35	0.648	0.542	**0.675**	0.630	0.641	0.659
40	0.633	0.502	**0.649**	0.610	0.627	0.634
45	**0.618**	0.475	0.617	0.592	0.613	0.610
50	0.596	0.452	**0.598**	0.574	0.597	0.595

(b) Novelty (NOV)

N	2CKG4ER	No-mastery	No-forgetting	No-student-feature	No-exercise-feature	ID-based only
5	0.989	0.982	**0.994**	0.988	0.989	0.991
10	0.988	0.984	**0.992**	0.986	0.985	0.988
15	0.985	0.986	**0.990**	0.986	0.985	0.986
20	0.986	0.984	**0.987**	0.986	0.985	0.986
25	0.985	0.984	**0.988**	0.987	0.987	0.985
30	0.986	0.981	**0.988**	0.987	0.987	0.986
35	0.987	0.979	**0.989**	0.987	0.988	0.987
40	0.987	0.978	**0.988**	0.987	0.987	0.987
45	0.987	0.976	**0.988**	0.986	0.987	0.987
50	0.985	0.976	**0.988**	0.987	0.987	0.986

Seed 2024; single-run values. No across-seed standard deviation is available.

Group 5: theta_m=0.8, theta_f=0.7, rec=10

6/6 models completed | completed | ranked across six models

(c) Ep, response threshold = 0.4

N	2CKG4ER	No-mastery	No-forgetting	No-student-feature	No-exercise-feature	ID-based only
5	-0.156	**0.087**	-0.157	0.032	-0.093	-0.162
10	0.062	**0.070**	-0.060	0.020	0.035	-0.008
15	0.091	0.187	0.179	0.040	**0.198**	0.007
20	0.118	0.238	**0.299**	0.098	0.147	0.173
25	0.340	0.027	0.227	0.125	0.242	**0.340**
30	**0.378**	-0.007	0.277	0.163	0.162	0.279
35	0.213	0.068	**0.376**	0.209	0.169	0.355
40	**0.285**	-0.009	0.085	0.259	0.221	0.232
45	0.181	0.059	0.346	0.272	0.208	**0.438**
50	0.058	0.119	0.128	**0.258**	0.229	0.098

(d) Ep, response threshold = 0.5

N	2CKG4ER	No-mastery	No-forgetting	No-student-feature	No-exercise-feature	ID-based only
5	-0.175	**0.083**	-0.164	0.002	-0.145	-0.191
10	0.060	**0.070**	-0.112	-0.020	0.025	-0.022
15	0.084	**0.187**	0.155	-0.002	0.173	0.005
20	0.112	0.238	**0.297**	0.064	0.142	0.173
25	0.338	0.027	0.135	0.101	0.233	**0.340**
30	**0.378**	-0.060	0.263	0.141	0.092	0.279
35	0.212	0.063	**0.373**	0.189	0.142	0.355
40	**0.285**	-0.080	0.084	0.231	0.202	0.232
45	0.115	0.034	0.338	0.259	0.201	**0.438**
50	0.043	0.113	0.124	**0.247**	0.225	0.098

Seed 2024; single-run values. No across-seed standard deviation is available.

Group 5: theta_m=0.8, theta_f=0.7, rec=10

6/6 models completed | completed | ranked across six models

(e) Ep, response threshold = 0.6

N	2CKG4ER	No-mastery	No-forgetting	No-student-feature	No-exercise-feature	ID-based only
5	-0.292	**-0.017**	-0.308	-0.110	-0.280	-0.281
10	0.021	**0.028**	-0.388	-0.179	-0.098	-0.181
15	-0.009	0.004	-0.218	-0.171	**0.070**	-0.275
20	-0.193	0.212	**0.249**	-0.110	0.110	0.061
25	0.219	0.013	0.117	-0.054	0.213	**0.300**
30	**0.293**	-0.092	0.257	0.010	0.038	0.143
35	0.088	0.049	0.195	0.072	0.008	**0.320**
40	**0.241**	-0.090	-0.083	0.141	0.147	0.223
45	0.092	-0.058	0.284	0.183	0.174	**0.425**
50	0.032	0.098	0.100	0.174	**0.197**	0.091

(f) Ep, response threshold = 0.7

N	2CKG4ER	No-mastery	No-forgetting	No-student-feature	No-exercise-feature	ID-based only
5	-0.399	**-0.179**	-0.360	-0.213	-0.375	-0.392
10	-0.278	**-0.206**	-0.427	-0.310	-0.249	-0.385
15	-0.282	-0.204	-0.265	-0.314	**-0.114**	-0.343
20	-0.310	**0.092**	0.045	-0.277	-0.150	-0.238
25	-0.034	-0.035	-0.122	-0.253	-0.159	**0.064**
30	-0.067	-0.249	**0.165**	-0.222	-0.263	-0.024
35	-0.247	0.017	**0.164**	-0.166	-0.266	-0.017
40	**-0.011**	-0.213	-0.209	-0.086	-0.117	-0.135
45	-0.218	-0.270	-0.048	-0.039	-0.037	**0.256**
50	-0.228	-0.102	-0.214	-0.045	**-0.004**	-0.179

Seed 2024; single-run values. No across-seed standard deviation is available.

Group 6: theta_m=0.8, theta_f=0.5, rec=10

6/6 models completed | completed | ranked across six models

(a) Adaptability (Ada)

N	2CKG4ER	No-mastery	No-forgetting	No-student-feature	No-exercise-feature	ID-based only
5	0.748	0.722	**0.788**	0.742	0.742	0.750
10	0.746	0.616	**0.766**	0.721	0.736	0.731
15	0.733	0.571	**0.740**	0.696	0.725	0.718
20	0.709	0.533	**0.718**	0.672	0.706	0.683
25	0.700	0.514	**0.703**	0.657	0.698	0.662
30	0.682	0.493	**0.701**	0.643	0.674	0.648
35	0.648	0.471	**0.673**	0.626	0.647	0.633
40	0.642	0.455	**0.643**	0.612	0.630	0.619
45	0.621	0.441	0.616	0.596	**0.622**	0.607
50	0.594	0.428	**0.603**	0.581	0.601	0.595

(b) Novelty (NOV)

N	2CKG4ER	No-mastery	No-forgetting	No-student-feature	No-exercise-feature	ID-based only
5	0.989	0.983	**0.994**	0.991	0.990	0.992
10	0.990	0.979	**0.992**	0.991	0.988	0.989
15	0.988	0.977	**0.989**	0.989	0.985	0.988
20	0.986	0.974	0.987	**0.989**	0.986	0.986
25	0.986	0.974	0.988	**0.988**	0.986	0.986
30	0.986	0.974	0.986	**0.988**	0.987	0.987
35	0.987	0.974	0.988	**0.988**	0.987	0.987
40	0.988	0.974	0.987	**0.988**	0.987	0.987
45	0.988	0.975	0.987	**0.989**	0.988	0.986
50	0.988	0.975	0.988	**0.989**	0.988	0.986

Seed 2024; single-run values. No across-seed standard deviation is available.

Group 6: theta_m=0.8, theta_f=0.5, rec=10

6/6 models completed | completed | ranked across six models

(c) Ep, response threshold = 0.4

N	2CKG4ER	No-mastery	No-forgetting	No-student-feature	No-exercise-feature	ID-based only
5	**0.005**	-0.029	-0.100	-0.043	-0.024	-0.071
10	**0.127**	0.028	-0.079	-0.000	-0.050	0.024
15	0.094	0.080	**0.147**	0.036	0.089	-0.002
20	0.154	0.132	**0.247**	0.088	0.161	0.079
25	**0.356**	0.150	0.288	0.139	0.230	0.127
30	0.150	0.129	0.202	0.168	0.157	**0.218**
35	0.220	0.093	**0.364**	0.190	0.043	0.185
40	0.174	0.117	**0.409**	0.168	0.041	0.164
45	**0.210**	0.071	0.058	0.193	0.170	0.160
50	0.178	0.093	-0.162	0.198	**0.298**	0.158

(d) Ep, response threshold = 0.5

N	2CKG4ER	No-mastery	No-forgetting	No-student-feature	No-exercise-feature	ID-based only
5	-0.123	-0.069	-0.110	**-0.064**	-0.083	-0.096
10	**0.117**	0.004	-0.143	-0.043	-0.058	0.007
15	0.093	0.070	**0.139**	-0.003	0.059	-0.026
20	0.139	0.130	**0.246**	0.050	0.137	0.021
25	**0.352**	0.150	0.288	0.102	0.225	0.082
30	0.148	0.129	**0.202**	0.141	0.155	0.199
35	0.220	0.092	**0.363**	0.167	-0.054	0.170
40	0.075	0.109	**0.409**	0.142	0.002	0.149
45	**0.193**	0.045	0.058	0.170	0.109	0.144
50	0.171	0.075	-0.162	0.176	**0.279**	0.136

Seed 2024; single-run values. No across-seed standard deviation is available.

Group 6: theta_m=0.8, theta_f=0.5, rec=10

6/6 models completed | completed | ranked across six models

(e) Ep, response threshold = 0.6

N	2CKG4ER	No-mastery	No-forgetting	No-student-feature	No-exercise-feature	ID-based only
5	-0.255	**-0.176**	-0.308	-0.198	-0.181	-0.204
10	**0.032**	-0.129	-0.410	-0.214	-0.099	-0.061
15	**-0.009**	-0.021	-0.177	-0.177	-0.098	-0.101
20	0.017	0.074	**0.176**	-0.115	0.045	-0.168
25	**0.308**	0.109	0.180	-0.070	0.181	-0.185
30	0.040	0.107	**0.126**	-0.008	0.115	-0.034
35	0.159	0.065	**0.168**	0.032	-0.139	0.036
40	0.052	0.085	**0.314**	0.021	-0.107	0.056
45	**0.186**	-0.003	-0.071	0.066	-0.033	0.076
50	0.167	0.021	-0.340	0.095	**0.211**	0.069

(f) Ep, response threshold = 0.7

N	2CKG4ER	No-mastery	No-forgetting	No-student-feature	No-exercise-feature	ID-based only
5	-0.313	**-0.284**	-0.403	-0.293	-0.302	-0.310
10	-0.277	-0.277	-0.448	-0.311	-0.278	**-0.232**
15	-0.294	**-0.207**	-0.214	-0.286	-0.289	-0.318
20	-0.243	-0.105	**0.086**	-0.237	-0.267	-0.296
25	-0.010	-0.085	**0.015**	-0.213	-0.188	-0.308
30	-0.250	-0.052	**0.007**	-0.193	-0.251	-0.196
35	-0.152	-0.065	**0.143**	-0.166	-0.346	-0.173
40	-0.276	-0.066	**0.182**	-0.192	-0.346	-0.203
45	-0.221	**-0.132**	-0.293	-0.170	-0.273	-0.212
50	-0.081	-0.140	-0.504	-0.153	**-0.027**	-0.205

Seed 2024; single-run values. No across-seed standard deviation is available.

Group 7: theta_m=0.8, theta_f=0.8, rec=30

6/6 models completed | completed | ranked across six models

(a) Adaptability (Ada)

N	2CKG4ER	No-mastery	No-forgetting	No-student-feature	No-exercise-feature	ID-based only
5	0.694	0.567	0.718	0.720	**0.725**	0.707
10	0.722	0.593	0.728	0.695	**0.731**	0.680
15	0.725	0.519	**0.729**	0.672	0.716	0.660
20	0.707	0.475	**0.726**	0.657	0.700	0.647
25	0.698	0.461	**0.723**	0.641	0.688	0.636
30	0.681	0.457	**0.722**	0.624	0.681	0.621
35	0.662	0.448	**0.717**	0.608	0.671	0.612
40	0.654	0.441	**0.715**	0.594	0.650	0.601
45	0.639	0.434	**0.708**	0.581	0.631	0.589
50	0.631	0.428	**0.697**	0.569	0.618	0.578

(b) Novelty (NOV)

N	2CKG4ER	No-mastery	No-forgetting	No-student-feature	No-exercise-feature	ID-based only
5	0.985	0.981	**0.991**	0.988	0.987	0.991
10	0.986	0.980	**0.991**	0.986	0.987	0.989
15	0.985	0.977	**0.992**	0.987	0.985	0.988
20	0.985	0.976	**0.991**	0.988	0.985	0.988
25	0.986	0.974	**0.991**	0.987	0.985	0.987
30	0.986	0.974	**0.990**	0.988	0.985	0.988
35	0.985	0.973	**0.990**	0.988	0.985	0.988
40	0.985	0.973	**0.989**	0.987	0.985	0.988
45	0.984	0.973	**0.989**	0.987	0.984	0.987
50	0.985	0.973	**0.989**	0.987	0.985	0.987

Seed 2024; single-run values. No across-seed standard deviation is available.

Group 7: theta_m=0.8, theta_f=0.8, rec=30

6/6 models completed | completed | ranked across six models

(c) Ep, response threshold = 0.4

N	2CKG4ER	No-mastery	No-forgetting	No-student-feature	No-exercise-feature	ID-based only
5	0.063	**0.083**	0.061	-0.019	-0.014	-0.019
10	-0.058	**0.054**	-0.052	0.013	-0.006	0.009
15	-0.013	**0.061**	-0.019	0.011	-0.044	0.055
20	-0.063	**0.074**	0.057	0.059	-0.089	0.053
25	-0.075	0.117	**0.118**	0.085	-0.016	0.086
30	-0.075	0.080	**0.165**	0.123	0.045	0.091
35	0.138	0.026	**0.218**	0.162	-0.012	0.122
40	**0.222**	0.008	0.165	0.163	0.138	0.156
45	0.116	0.064	**0.261**	0.166	0.194	0.168
50	0.235	-0.007	0.240	0.163	**0.240**	0.196

(d) Ep, response threshold = 0.5

N	2CKG4ER	No-mastery	No-forgetting	No-student-feature	No-exercise-feature	ID-based only
5	0.055	**0.069**	0.057	-0.039	-0.020	-0.032
10	-0.095	**0.039**	-0.053	-0.022	-0.012	-0.013
15	-0.108	**0.038**	-0.019	-0.037	-0.068	0.028
20	-0.114	**0.057**	0.056	0.010	-0.102	0.027
25	-0.089	0.090	**0.118**	0.048	-0.059	0.060
30	-0.126	0.062	**0.165**	0.092	0.016	0.063
35	0.099	-0.008	**0.218**	0.137	-0.029	0.100
40	**0.209**	-0.014	0.165	0.143	0.130	0.136
45	0.106	0.048	**0.244**	0.145	0.186	0.146
50	0.221	-0.031	0.230	0.132	**0.236**	0.174

Seed 2024; single-run values. No across-seed standard deviation is available.

Group 7: theta_m=0.8, theta_f=0.8, rec=30

6/6 models completed | completed | ranked across six models

(e) Ep, response threshold = 0.6

N	2CKG4ER	No-mastery	No-forgetting	No-student-feature	No-exercise-feature	ID-based only
5	**0.039**	0.023	-0.034	-0.111	-0.081	-0.122
10	-0.240	**-0.030**	-0.194	-0.130	-0.085	-0.140
15	-0.318	**-0.075**	-0.290	-0.163	-0.190	-0.107
20	-0.408	**-0.068**	-0.306	-0.132	-0.335	-0.119
25	-0.463	**0.035**	-0.280	-0.105	-0.355	-0.095
30	-0.432	**-0.007**	-0.117	-0.078	-0.318	-0.081
35	-0.142	-0.079	**0.134**	-0.022	-0.410	-0.040
40	0.091	-0.082	**0.150**	0.004	-0.037	0.028
45	0.039	-0.021	**0.187**	0.025	0.110	0.046
50	0.179	-0.098	0.147	0.033	**0.179**	0.079

(f) Ep, response threshold = 0.7

N	2CKG4ER	No-mastery	No-forgetting	No-student-feature	No-exercise-feature	ID-based only
5	**-0.010**	-0.112	-0.149	-0.175	-0.203	-0.215
10	-0.352	**-0.208**	-0.352	-0.233	-0.314	-0.264
15	-0.371	**-0.193**	-0.386	-0.277	-0.387	-0.259
20	-0.455	**-0.187**	-0.371	-0.259	-0.456	-0.279
25	-0.485	**-0.057**	-0.325	-0.242	-0.434	-0.258
30	-0.447	**-0.117**	-0.251	-0.220	-0.370	-0.264
35	-0.263	-0.181	**-0.163**	-0.190	-0.431	-0.231
40	**-0.134**	-0.239	-0.251	-0.175	-0.217	-0.178
45	-0.232	-0.157	-0.144	-0.163	**-0.115**	-0.166
50	-0.046	-0.228	-0.175	-0.147	**0.016**	-0.137

Seed 2024; single-run values. No across-seed standard deviation is available.

Group 8: theta_m=0.8, theta_f=0.8, rec=50

6/6 models completed | completed | ranked across six models

(a) Adaptability (Ada)

N	2CKG4ER	No-mastery	No-forgetting	No-student-feature	No-exercise-feature	ID-based only
5	**0.774**	0.468	0.699	0.738	0.767	0.678
10	**0.746**	0.416	0.699	0.715	0.746	0.661
15	**0.736**	0.421	0.699	0.697	0.732	0.649
20	**0.723**	0.417	0.699	0.678	0.715	0.635
25	0.692	0.426	**0.702**	0.668	0.694	0.625
30	0.670	0.423	**0.708**	0.656	0.667	0.614
35	0.656	0.430	**0.711**	0.644	0.651	0.607
40	0.649	0.422	**0.715**	0.634	0.645	0.596
45	0.643	0.413	**0.716**	0.621	0.641	0.587
50	0.641	0.402	**0.715**	0.610	0.637	0.579

(b) Novelty (NOV)

N	2CKG4ER	No-mastery	No-forgetting	No-student-feature	No-exercise-feature	ID-based only
5	0.990	0.975	0.991	0.990	**0.991**	0.981
10	0.988	0.974	**0.991**	0.988	0.990	0.982
15	0.987	0.975	**0.991**	0.989	0.986	0.984
20	0.986	0.975	**0.991**	0.988	0.985	0.985
25	0.985	0.976	**0.991**	0.988	0.985	0.984
30	0.985	0.975	**0.991**	0.988	0.984	0.984
35	0.984	0.976	**0.991**	0.988	0.983	0.984
40	0.984	0.976	**0.991**	0.988	0.983	0.983
45	0.984	0.975	**0.991**	0.988	0.983	0.984
50	0.985	0.975	**0.990**	0.988	0.984	0.984

Seed 2024; single-run values. No across-seed standard deviation is available.

Group 8: theta_m=0.8, theta_f=0.8, rec=50

6/6 models completed | completed | ranked across six models

(c) Ep, response threshold = 0.4

N	2CKG4ER	No-mastery	No-forgetting	No-student-feature	No-exercise-feature	ID-based only
5	-0.132	0.023	**0.169**	-0.077	-0.135	0.040
10	0.107	0.068	**0.233**	-0.012	0.051	0.064
15	0.033	0.124	**0.246**	0.043	0.031	0.080
20	-0.032	0.116	**0.263**	0.082	-0.061	0.089
25	-0.182	0.126	**0.161**	0.088	-0.112	0.068
30	0.018	0.059	-0.083	**0.103**	-0.002	0.032
35	0.193	0.004	0.010	0.126	**0.200**	0.088
40	0.224	0.009	0.023	0.143	**0.245**	0.098
45	0.252	-0.029	0.100	0.138	**0.260**	0.095
50	**0.224**	0.025	0.195	0.145	0.202	0.071

(d) Ep, response threshold = 0.5

N	2CKG4ER	No-mastery	No-forgetting	No-student-feature	No-exercise-feature	ID-based only
5	-0.186	0.006	**0.166**	-0.095	-0.162	0.025
10	0.074	0.046	**0.231**	-0.060	0.037	0.050
15	0.024	0.111	**0.245**	-0.002	0.027	0.069
20	-0.033	0.108	**0.263**	0.043	-0.062	0.075
25	-0.226	0.122	**0.160**	0.063	-0.124	0.044
30	-0.102	0.050	-0.101	**0.085**	-0.030	-0.001
35	0.167	-0.023	-0.004	0.105	**0.190**	0.053
40	0.211	-0.016	0.021	0.122	**0.241**	0.056
45	0.244	-0.055	0.099	0.107	**0.255**	0.057
50	**0.218**	0.007	0.195	0.112	0.195	0.032

Seed 2024; single-run values. No across-seed standard deviation is available.

Group 8: theta_m=0.8, theta_f=0.8, rec=50

6/6 models completed | completed | ranked across six models

(e) Ep, response threshold = 0.6

N	2CKG4ER	No-mastery	No-forgetting	No-student-feature	No-exercise-feature	ID-based only
5	-0.318	-0.053	**0.112**	-0.201	-0.281	-0.030
10	-0.037	-0.050	**0.174**	-0.192	-0.100	-0.038
15	-0.059	0.083	**0.204**	-0.155	-0.103	-0.032
20	-0.150	0.075	**0.228**	-0.115	-0.226	-0.031
25	-0.409	0.083	**0.124**	-0.102	-0.371	-0.073
30	-0.302	**-0.027**	-0.253	-0.079	-0.317	-0.128
35	0.052	-0.093	-0.353	-0.059	**0.066**	-0.095
40	0.129	-0.065	-0.393	-0.032	**0.177**	-0.093
45	0.191	-0.121	-0.338	-0.027	**0.210**	-0.086
50	**0.172**	-0.053	-0.291	-0.021	0.136	-0.111

(f) Ep, response threshold = 0.7

N	2CKG4ER	No-mastery	No-forgetting	No-student-feature	No-exercise-feature	ID-based only
5	-0.388	-0.160	**-0.009**	-0.268	-0.378	-0.098
10	-0.103	-0.148	**0.011**	-0.295	-0.273	-0.140
15	-0.273	-0.017	**0.006**	-0.266	-0.259	-0.175
20	-0.332	-0.029	**0.010**	-0.257	-0.424	-0.203
25	-0.451	**-0.024**	-0.178	-0.249	-0.446	-0.257
30	-0.344	**-0.184**	-0.455	-0.248	-0.392	-0.302
35	**-0.016**	-0.262	-0.414	-0.226	-0.104	-0.261
40	**0.081**	-0.265	-0.409	-0.219	0.072	-0.262
45	**0.137**	-0.284	-0.345	-0.217	0.136	-0.271
50	**0.121**	-0.246	-0.295	-0.210	0.038	-0.292

Seed 2024; single-run values. No across-seed standard deviation is available.